UNIVERSITÀ DI TRENTO

Introduction

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Welcome to AdvancedHCI!

- Who I am

  Giovanna Varni, Associate Professor @DISI

  Office: 208, Povo 1

  Office hours: please, send me an email to fix a time slot!

- Advanced HCI *....that is???*
  - So **MANY** things!
  - Focus on:
    - **Multimodality**
    - **Social Signal Processing (SSP) for Social AI (SAI):**

      Two scientific domains aimed at endowing machines with social intelligence (making sense of social interaction as humans do: perception (SSP) and action (SAI))

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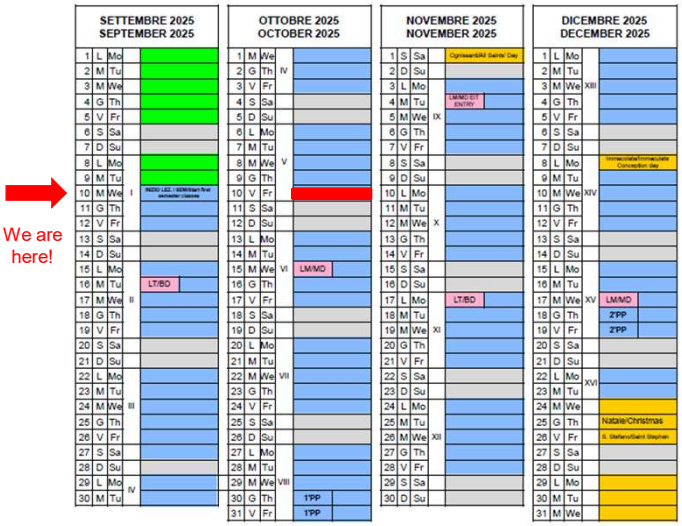
About the course #1



We are here!

Course schedule:
Tuesday, 8:30pm – 10:30 am
Friday, 2:30pm – 4:30pm

Course duration: 48h

No lecture on October 10th

End of the semester

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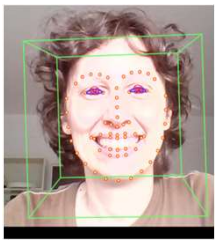
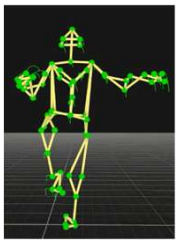
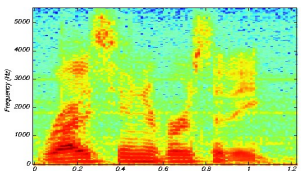
About the course #2

Topics:

- Short introduction to HCI and recap on basic bricks
- Multimodality and Social Signal Processing for Social AI
- Devices and technologies for multimodal interaction
- Computational approaches to analyze and interpret human social behavior: facial expression, gaze, gesture, voice
- Hints on Multimodal Machine Learning (MML)
- Examples and evaluation







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OK, but to do what?

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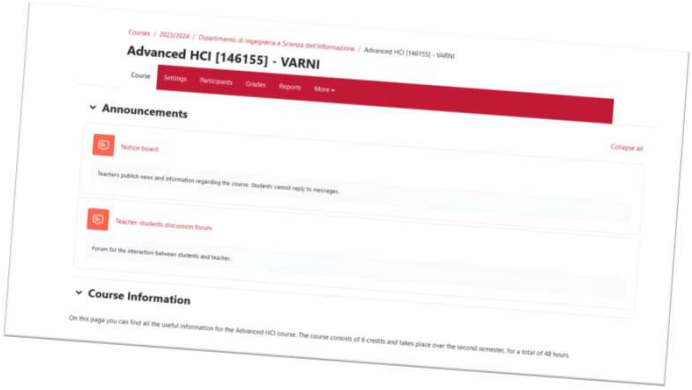
About the course #3

- Check out the Moodle of the course for news and materials!



- If you do not have access, please:
  - verify with the student office; and
  - send me an email

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About the course #4

Nice-to-have requirements:

- Some knowledge on Python / C / C++ programming
- Some maths and possibly basic machine learning / signal processing
- Be curious about science and data
- To want to have fun!

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About the course #5

Books and references:

- We do not use any book in particular
- At the end of each topic, (some) references to scientific papers and or books are provided
- We will use some tools with their tutorials

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Exam step by step #1

- The exam consists of a project and an oral test
- 5 exam calls per year

- About the project (60%):
  - It is structured in several parts / presentations
  - It will be partially developed during the lectures

- About the oral test (40%):
  - Project presentation and discussion with the teacher
  - Questions on course topics

3 Homeworks giving 1 extra point on the final mark!

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Exam step by step #2

- You *must*:
  - register for the exam 3 days before through the ESSE3 system;
  - upload the final version of your project 3 days before the exam

- If you are not satisfy with your mark, try again and re-upload your project with improvements, if needed

- If you repeat the exam, the new mark overwrites the old one

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Project Topic a.y. 2025-2026

Physical, mental, societal wellbeing

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Going further…

If you have a particular interest in working on the topics covered by the course or on related topics, I am available to supervise:

- Project course (6 ECTS)
- Research project (12 ECTS)
- Internship and Master's thesis

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Before to begin…

- Let's talk ☺. What about:
  - Your background
  - Your skills and interests
  - Why are you here? So, what do you expect from this course?
- Who wants to start?

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Any futher question?
...Let's start!

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HCI essentials in a nutshell

- *"Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them"*

  (Association Computation Machinery, ACM)

- The HCI paradox:
  - Success is measured by what cannot be seen
  - Making the complex simple

- HCI is a transdisciplinary field:
  - Computer Science (Programming, Computer Vision, Data Science, Security…), Cognitive Sciences, Human Factors, Psychology and much more!

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HCI essentials in a nutshell

- HCI ingredients:
  - Human beings (single user, teams, society)
  - Machines (PC, embedded systems, robots, virtual humans)
  - Some task to solve to achieve a common goal

- A key point:

Human beings and machines interact together over time!

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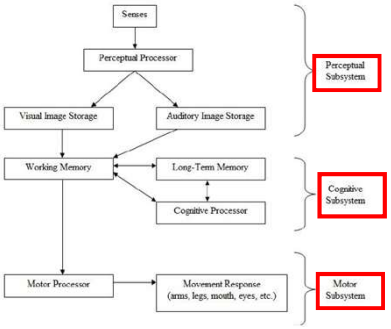
HCI essentials in a nutshell

Human



Card, Moran, Newell, The psychology of human computer interaction, 1983

Computer

Input devices:
- Keyboard, mouse
- Touch surfaces
- Sensing systems

Ouput devices:
- Screen
- Artificial agents
- AR/VR headsets

Processing:
- CPU
- GPU

Task

What we want / need to do to achieve goals

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HCI essentials in a nutshell

Concept	View of interaction	Key phenomena and constructs	Good interaction	Example support for evaluation and design
Dialogue	a cyclic process of communication acts and their interpretations	mappings between UI and intentions; feedback from the UI; turn taking	understandable; simple, natural; direct	methods/concepts for guessability, feedback, mapping; walkthroughs
Transmission	a sender sending a message over a noisy channel	messages (bits); sender and receiver; noisy channels	maximum throughput of information	metrics and models of user performance
Tool use	a human that uses tools to manipulate and act in the world	mediation by tools; directness of acting in the world; activity as a unit of analysis	useful and transparent tools; amplification of human capabilities	compatibility in instrumental interaction; break down analysis
Optimal behavior	adapting behavior to goals, task, UI, and capabilities	rationality; constraints; preferences; utility; strategies	improves or reaches maximum or satisfactory utility	models of choice, foraging, and adaptation
Embodiment	acting and being in situations of a material and social world	intentionality; context; coupling	provides resources for and supports fluent participation in the world	studies in the wild; thick description
Experience	an ongoing stream of expectations, feelings, memories	non-utilitarian quality; expectations; emotion	satisfies psychological needs; motivating	metrics of user experience; experience design methods
Control	interactive minimization of error against some reference	feedforward; feedback; reference; system; dynamics	rapid and stable convergence to target state	executable simulations of interactive control tasks

Kasper Hornbæk & Antti Oulasvirta. What Is Interaction? In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*

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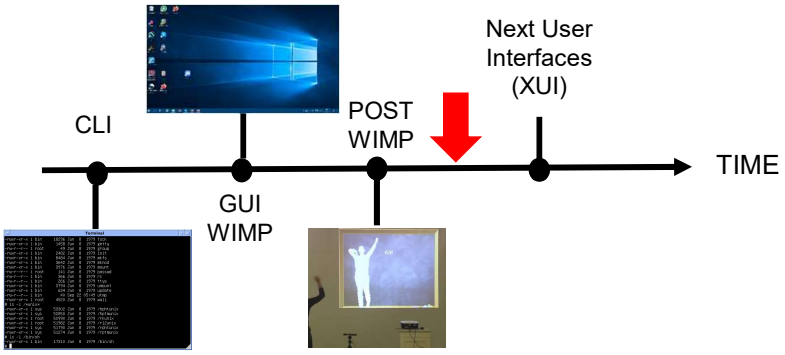
Long story short, here's the history of HCI

- A punctuated equilibrium



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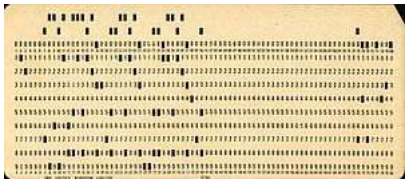
Once upon a time…

- The first era (pre-CLI): early 1950s and 1960s
  - Computers used in batch mode with punched-card input and line-printer output
  - No user interfaces (not interactive users)

Source Wikimedia: ArnoldReinhold, CC BY-SA 3.0



Source Wikipedia: Pete Birkinshaw, CC BY 2.0

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Once upon a time…

- The second era: early 1960s through early 1980s
  - timesharing on mainframes and minicomputers using mechanical or “glass” teletypes (alphanumeric displays)
  - users could interact with the computer by typing in commands with parameters
  - Command-line Interfaces (CLI)

Source Wikimedia:Gorthmog, CC BY-SA 4.0

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CLI

PROS

- Knowing commands allows user to go fast
- Low use of RAM and CPU
- Low resolution monitor can be used

CONS

- Confusing for novices
- Spelling command errors result in fail and often in starting from zero
- Larger number of commands need to be learned
- Impossible to guess what an instruction might be

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From the desk…

- The third era: from 1970 - to now
  - from the beginning of 1970s attempt to make computers more user-friendly by making them resemble the common workplace of the time (with papers, folders, and a desktop)

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…to the paper paradigm (1968)

- It consists of, usually, black text on a white background, files within folders, and a "desktop“
- It was created by many individuals and organizations, such as Douglas Engelbart, Xerox PARC, and Apple Computer
- It is the common paradigm used by most modern OS

The Mother of All Demos (1968)

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…to the desktop metaphor

- Alan Kay at Xerox PARC

  The computer interface is a conceptual metaphor of a writing desk

source Wikimedia: Sailko, CC BY 3.0

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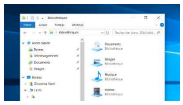
Implementing the desktop metaphor

- The desktop metaphor is a set of unifying concepts used by GUIs to help users interact more easily with the computer
- A traditional way to do that: the **WIMP paradigm (Wilberts,1980)**
- “Point and click” interfaces

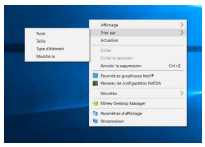
W.I.M.P.

Window



Icon

Menu



Pointer

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WIMP

PROS

- Simple to explore / use especially for novices
- Do not have to learn complicated commands
- Benefits of WYSIWYG
- Exchange data between different software applications
- Suitable for multitasking environment

CONS

- More RAM / CPU and hard disk space
- Slow for experienced programmers
- “drowning in functionality”
- Limited input channels
- Well suited only for 2D applications
- Designed for a single desktop user

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A big issue…

WIMPs are the perfect interfaces only for creatures with a single eye, one or more single-jointed fingers… (Buxton)

BUT

Humans are multimodal by nature!

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Where we are…

- The fourth era: from 1997 to now
  - Van Dam defined as post-WIMP the interfaces *"containing at least one interaction technique not dependent on classical 2D widgets such as menus and icons"*
  - That is:
    - At least one non-classical inputs: speech, gesture, touch
    - Mostly process parallel input streams
    - Can support multiple users (human beings as well as machines)

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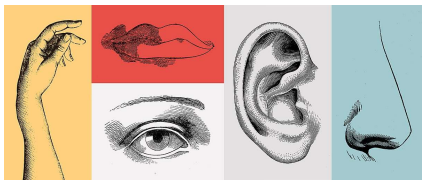
Basic terminology

- Sensor modality: sensory channel through which info is perceived / sent



https://healthmatters.nyp.org/the-science-behind-love-how-your-brain-and-five-senses-help-you-fall-in-love/

- Multimodal so refers to the **integration** of info coming from / starting from different sensory channels for extracting / conveying **high level meaning**
- Multimedia vs Multimodal:
  - Multimedia systems **convey** signals of different kinds (e.g. a ppt presentation with audio / visual clip)
  - Multimodal systems **interpret** signals from various sensory modalities

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Theoretical foundations of multimodal interaction

	Key implications
Gestalt holistic systems-level view of perception, which emphasizes self-organization of perceptual experience into meaningful wholes, rather than analyzing discrete elements as isolates	When errors or task difficulty increase, users adapt the co-timing of their individual signals to fortify the whole multimodal construction
Working memory Ability to store information temporarly (few secs) in mind before to be consolidated into long-term memory	Performances improve when a system combines different modalities supporting complementary information processing in separate brain regions simoultaneously
Activity theory Activity and consciousness are dynamically interrelated	Multimodal systems stimulates cognition producing a deeper and more elaborated learning and improving long-term memory

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Humans are multimodal by nature!

Quiz time!

Inspired by the talk of L.P. Morency: Modeling Human Communication Dynamics

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Human Multimodal Communication

INTERVIEWER: "…Guy Kweney is the editor of the technology website News Wireless. Hello, good morning to you."

INTERVIEWEE: "Good morning."

INTERVIEWER: "Were you surprised by this verdict today?"

INTERVIEWEE: "I'm very surprised to see this verdict to come on me. Because I was not expecting that. When I came they told me something else and I'm coming. And they told me something else. Big surprise any way."

INTERVIEWER: "A big surprise..."

INTERVIEWEE: "Exactly."

INTERVIEWER: "Yeah yeah. With regard to the cost that is involved. Do you think more people will be downloading online?"

INTERVIEWEE: "Actually if you can go everywhere, you gonna see people downloading through the internet and the websites. [...]

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Human Multimodal Communication

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Human Multimodal Communication

Something more yet

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Romeo meets Juliet

Visual:

- Gesture & Body language
- Eye contact
- Facial expressions

Auditory:

- Prosody
- Vocal expressions

Verbal:

- Lexicon
- Syntax
- Pragmatics

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Human Communication

- Communication is…

…the use of verbal and nonverbal messages to create ongoing influence

…a process of sorting, selecting and sending symbols in such a way to help a receiver to find in his or her own mind a meaning similar to that intended by a sender

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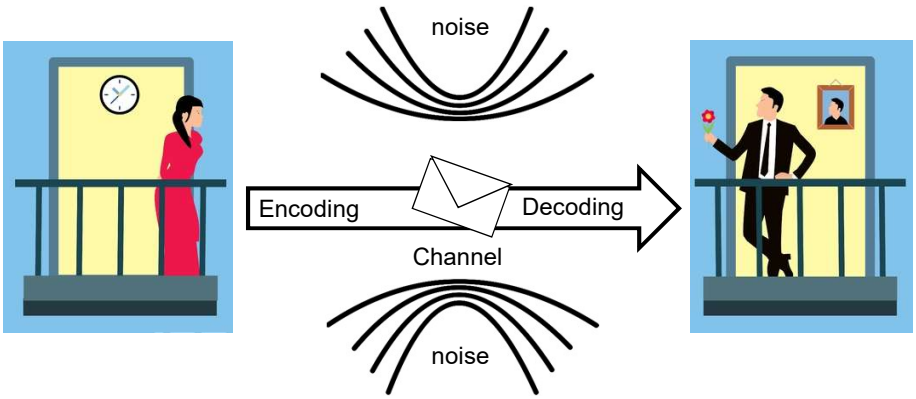
Human Communication Models

- Linear Model (Shannon & Weaver, 1948)



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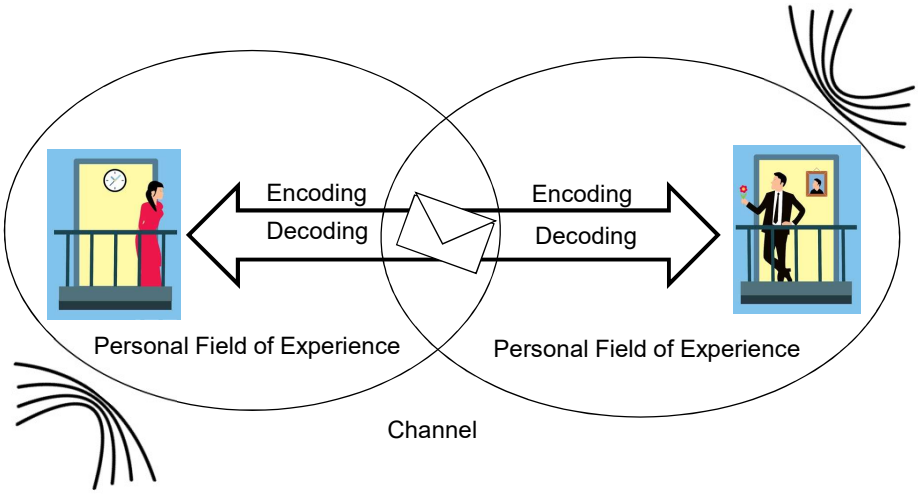
Human Communication Models

- Transactional Model (Barnlund, 1970)



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What about Human-Machine Communication (interaction)?

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From HCI: Multimodal Systems

Photo by Nadin Mario on Unsplash

- Perception
- Semantics
- Affects, moods
- Intentions, goals

- In 80'-90': developing systems capable of integrating and interpreting multiple modalities of data simultaneously, in a manner akin to human cognitive processing. Not much emphasis on intelligence
- In 2000 the W3C proposes a general framework to standardize multimodal systems. There's some sign of intelligence

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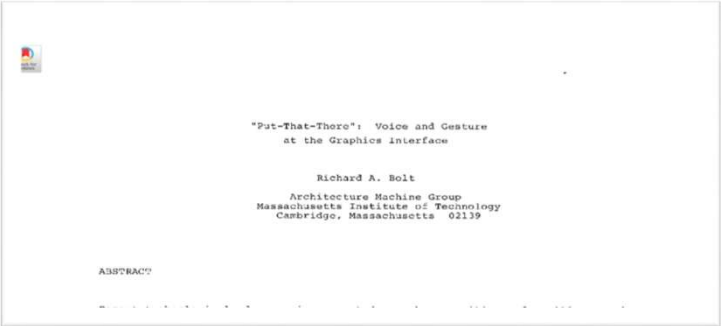
The first multimodal system

- Put that there (Bolt, 1980)



"Put-That-There": Voice and Gesture
at the Graphics Interface

Richard A. Bolt

Architecture Machine Group
Massachusetts Institute of Technology
Cambridge, Massachusetts 02139

ABSTRACT

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What is a multimodal system?

- Multimodal systems are *"systems that support a user communicating with an application by using different modalities such as voice (in a human language), gesture, handwriting, typing, audio-visual speech, etc."*
  (W3C Multimodal Interaction Working Group, 2003)

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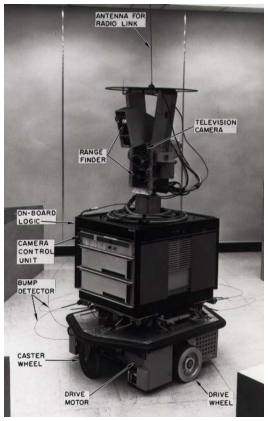
From AI: Artificial Intelligent Systems / Agents

- From 50'- 90': developing systems capable of intelligent (later on, autonomous) behavior within an environment



source Wikimedia: SRI International, CC BY-SA 3.0

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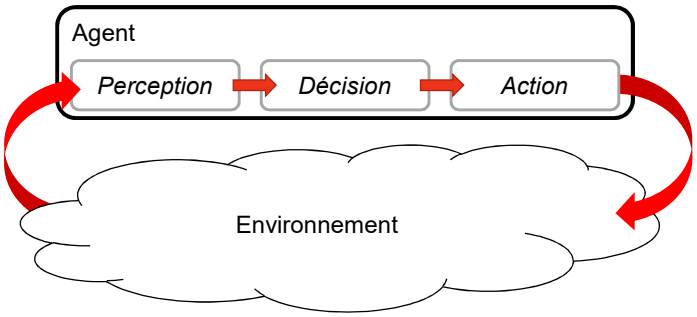
Perception – Decision - Action Loop



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What changes from 2010 onwards?

- Artificial systems / agents begin to require multimodal infrastructures, while multimodal systems start to be enriched with cognitive capabilities
- Thus, nowadays they are the two sides of the same coin!

Photo by Possessed Photography on Unsplash

- Multimodal understanding
- ASR, TTS, Gesture recognition…
- Behavioural analysis
- Planning

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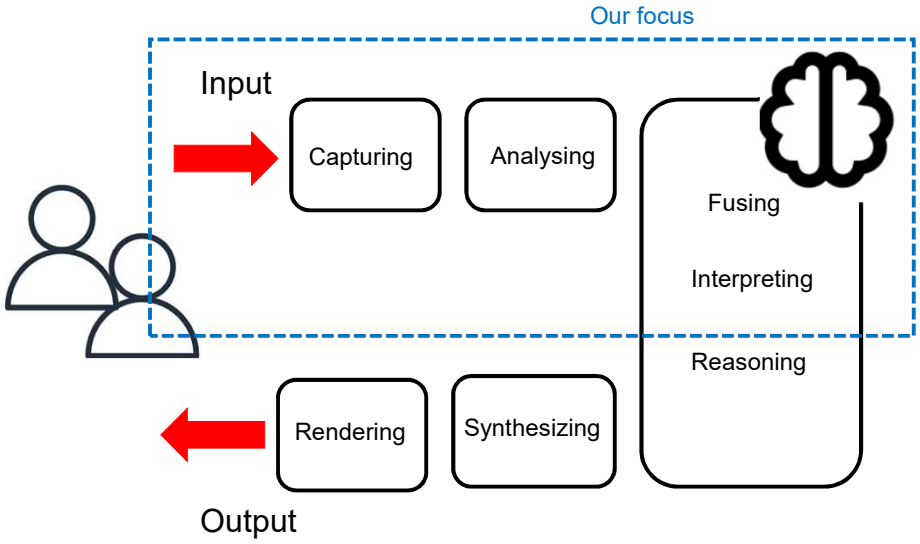
From a raw architectural point of view



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Input & output

- The Input / output may be classified as:
  - *Sequential:* it is received by / sent on a single modality, though that modality can change over time
  - *Simultaneous:* it is received by / sent on multiple modalities, and treated separately
  - *Composite:* it is received by / sent on multiple modalities at the same time and treated as a single, integrated "composite" input
- Simultaneous and composite input / output may be *conflicting*, in that the interpretations of the input may not be consistent (e.g., the user says "yes" but clicks on "no")

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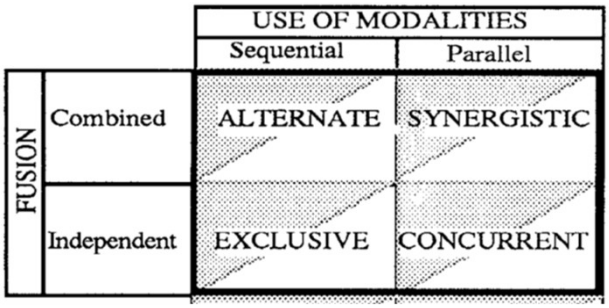
Input & output

- Nigay & Coutaz (1993) classified multimodal systems depending on:
  - Fusion method (combined or independent)
  - Use of modalities (sequential or parallel)



Picture from (Nigay & Coutaz, 1993)

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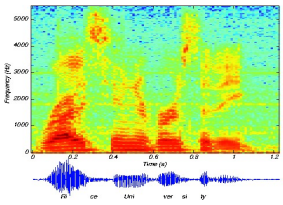
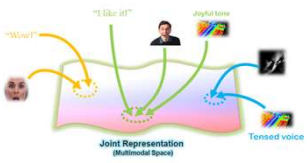
Capturing, Analysing & co.

- Capturing and analysis depend on the modality

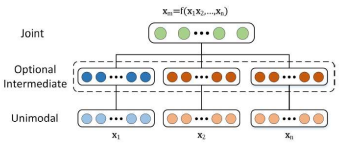


Picture from (Cao et al., 2016)

- Fusion and integration of multimodal data is part of Multimodal Machine Learning



Picture from Tutorial on Multimodal Machine Learning (L.-P. Morency)



Picture from (Baltrusaitis et al.,2019)

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Relevant Conferences – ICMI 2025

27th ACM International Conference on Multimodal Interaction
(13-17 Oct 2025)
Safe and responsible multimodal interactions

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Relevant Conferences – FG 2025

Home Dates Participate Organizers Keynotes Registration Sponsors Contacts

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Relevant Conferences – ACII2025

13th International Conference on Affective Computing and Intelligent Interaction (ACII 2025)-Canberra, Australia, 8-11 Oct 2025

Home Calls Attend Authors Program Important Dates People Contacts

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Relevant Conferences – IUI2025

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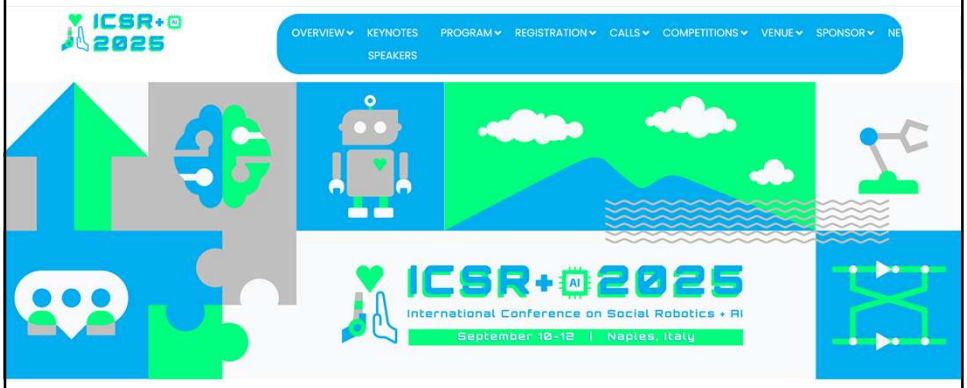
Relevant Conferences – ICSR2025



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Homework #1

- Visit the websites of some previous editions of these conferences
- Have a look at least at 3 papers (you can have access using your unitn account). Focus **only** on the topics: read titles and abstract
- Write a 1-page document reporting your likes and dislikes, your feelings and your questions on the topics interested you

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